

CENG428 Neural Networks

Homework 1 & 2

In this homework, you will design and implement a neural network model and train it using the provided training dataset. You are required to split the dataset into **training** and **validation** subsets and report validation performance during training. You will work *alone* or as a *2-student team*. Only one student will submit the Jupyter Notebook which includes all the code, visuals, and comments.

Your submission must be a single Jupyter Notebook consisting of two main parts:

1. **Training Phase** → In this part, you will:
 - define your model architecture,
 - train the network using the training data,
 - monitor and report training and validation performance
 - loss and performance values for all outputs
2. **Testing Phase** → In this part, you will:
 - load a separate test dataset,
 - evaluate your trained model,
 - report the final performance metrics.

A sample test dataset will be provided so that you can develop and verify your notebook. However, note that **the actual test data used for grading will be different**, so your solution should generalize well and must not rely on overfitting to the provided sample test set.

Network Input

For each input there will be n data points as (x_i, y_i) coordinates $0 \leq i < n$. Note that input size (n) and point order may vary across samples. Your model must handle this and produce a fixed-size representation. Data will be provided as NumPy arrays (see [numpy.load](#)).

Network Output

Your network must produce **7 outputs** for each input:

1. **Circle existence (1 value)**
 - Binary output indicating whether a circle exists.
2. **Thickness class (3 values)**
 - Three scores corresponding to: **thin**, **medium**, or **thick**.
3. **Circle parameters (3 values)**
 - (x, y, r) : center coordinates and radius.

Requirements

- Use PyTorch and automatic differentiation.
- Design your own neural network architecture.
- Implement a complete training pipeline.
- Use a proper multi-task loss. Thickness and circle parameters are only meaningful when a circle exists and should be ignored in the loss otherwise.

Do not use approaches we did not cover in the courses such as CNN, Transformer, ViT etc.

Deadline: 27 March 2026, 23:59